

PRANITA KHANAL

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Education

University of Texas at Arlington

Bachelor of Science in Computer Science, Minor in Mathematics

Arlington, TX

Aug 2023 – May 2027

Work Experience

Department of Physics

The University of Texas at Arlington

Lab Technician (under Dr. Madhab Pokhrel)

Jan 2024 – Aug 2026

- Modified lab manuals, create new experiments, and manage inventory logistics by conducting vendor cost-benefit analyses to optimize the departmental budget.
- Managed the installation and maintenance of lab computers and digital interfaces; lead the integration of PASCO Capstone and Data Studio software to modernize data acquisition and streamline experiment workflows.
- Utilized AutoCAD to engineer custom mechanical components for Mechanics labs; perform hardware repairs, including precision soldering and circuitry to maintain assets.

CyberGuard Lab

The University of Texas at Arlington

Research Assistant (under Dr. Faysal Hossain Shezan)

May 2026 – Present

- Achieved up to 3.36x inflation in retained visual tokens by designing and implementing the Budget Inflation Attack (BIA), a novel adversarial method exposing a resource-exhaustion vulnerability in adaptive visual token compression for Vision-Language Models, while leaving task accuracy unchanged and evading standard integrity-based detection.
- Benchmarked BIA across POPE, MMB, MME, and TextVQA datasets on the TACC HPC cluster, confirming consistent cost amplification (compute, energy, latency) with task accuracy left essentially unchanged, exposing adaptive token compression as a resource-exhaustion surface.

Projects

RAG PDF Chatbot

Dec 2025

Python, CSS, HTML, JS

pdf-chatbot-frontend-steel.vercel.app

- Built a full-stack AI chatbot with FastAPI backend and JS frontend using RAG architecture with Pinecone vector database and Google Gemini 2.5 API.
- Deployed containerized microservices on Hugging Face Spaces and Vercel with a CI/CD pipeline via GitHub for automated deployments.
- Engineered semantic search using Sentence Transformers for local embeddings, optimizing API costs while maintaining response accuracy.

Neural Network Library

Mar 2026

Python, NumPy

NNL@github

- Built a modular neural network library from scratch supporting arbitrary depth/width with linear layers, ReLU, sigmoid, and full backpropagation.
- Applied library to NYC taxi trip duration prediction, performing feature engineering, normalization, and hyperparameter search across 3+ model configurations.
- Validated XOR classification with a custom 2-layer network, comparing sigmoid vs. tanh activations to analyze convergence behavior.

Technical Skills

Languages: Python, Java, C, JavaScript, C++, SQL, PHP, Assembly, R, HTML/CSS

Frameworks & Libraries: Spring Boot, Spring Data JPA, Apache Kafka, FastAPI, React.js, RESTful API

AI/ML Tools: Scikit-learn, Pandas, NumPy, Matplotlib, Sentence Transformer

Data Science & Analytics: Data Analysis, Data Modeling, Data Visualization

DevOps & Tools: Docker, Git/GitHub, CI/CD Pipeline, Maven, Linux/UNIX, API Integration

Leadership / Extracurricular

Vice President & Treasurer

Summer, Fall 2026

Nepalese Students' Association

University of Texas at Arlington

- Represent and advocate for 300+ Nepali students at UTA, organizing 10+ cultural events and managing chapter finances to support community engagement and organizational growth.

Project Instructor

2018 – 2020

UK SARAID, The Japan Foundation

Kathmandu, Nepal

- Coordinated SMART Camp and instructed the HANDS-selected CAC project "SMART 2019" for the Kathmandu district with 150 participants; assisted in SMART's international COVID-19 session with cross-country research teams.